

Requirement Analysis - Technology Stack

Parameter	Detail
Date	18 June 2026
Team ID	Book-a-Doctor Platform
Project Name	Book a Doctor
Maximum Marks	4 Marks

Technical Architecture Diagram

The "Book a Doctor" platform utilizes a **3-tier Architecture** with MVC layout patterns separating user interaction, business controllers, and data storage.



Table-1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	The frontend UI which patient, doctor, and admin users interact with. Responsive web app with dark/light mode toggle.	React 18, Bootstrap 5, Vanilla CSS
2	Routing & Client State	Handles navigation, views loading, and client-side authentication checks.	React Router DOM v6, Axios API instance
3	Application Logic (API)	Backend REST framework managing authentication, bookings, vetting queue, and notifications.	Node.js, Express.js (MVC organization)
4	Database	Persists user metadata, doctor profile specifications, scheduling lists, and notification histories.	MongoDB Atlas, Mongoose ODM
5	Security Components	Module for encrypting passwords and signing authorization credentials.	Bcrypt.js, JsonWebToken (JWT)
6	File Upload / Storage	Logic for handling patient health reports upload and validator filters.	Multer middleware, local disk storage
7	Automation & Testing	Automatic seeding script and backend integration testing suites.	Jest, Supertest

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Core development relies on open-source Node and React packages, which prevents vendor lock-in.	React, Express, Mongoose, Axios
2	Security Implementations	Encryption of credentials, protected cookie sessions, express-validator sanitizers, and JWT authorization guards.	Bcrypt (10 rounds), HttpOnly cookie tokens, JWT, express-validator guards
3	Scalable Architecture	Modular directory structure dividing routers, controllers, middlewares, models, and validators for simple microservice scaling.	Express Router modularization, MVC structure
4	Availability	Deployment targets cloud services featuring auto-recovery and scaling groups.	Render Hosting, MongoDB Atlas clusters
5	Performance	Fast database lookups using indexing, and minimized network footprints by using light JSON models.	MongoDB single-field indices on specialty and status